5. If the period of revolution of an artificial satellite just above the earth's surface is T and density of earth is ρ then ρT^2

(1) Is universal constant whose value is $\frac{3\pi}{G}$

(2) Is universal constant whose value is $\frac{3\pi}{2G}$

(3) Is proportional to square of radius of earth R

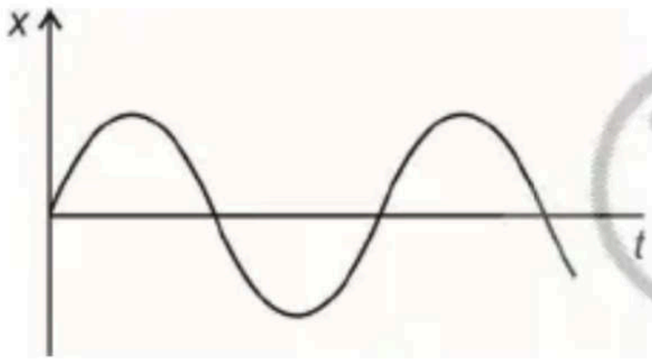
(4) Is proportional to radius of earth R

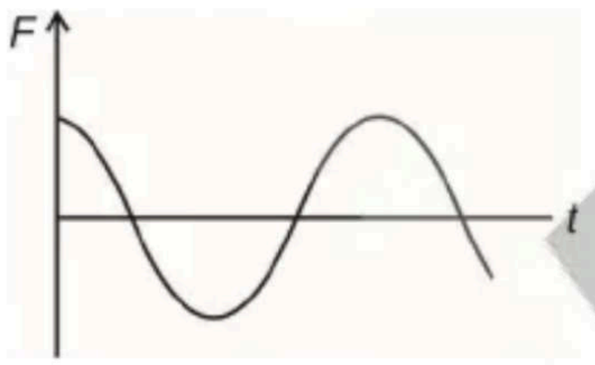
6. A uniform spherical shell gradually shrink maintaining its shape. The gravitational potential at its centre will (Mass of the shell remains same)

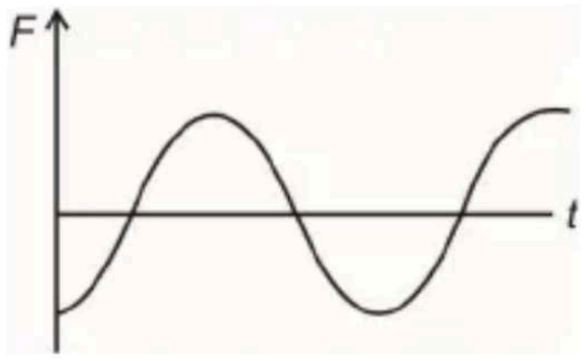
(1) Increase (2) Decrease

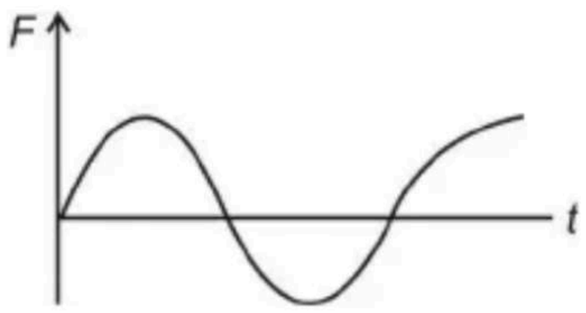
(3) Remain same (4) None of these

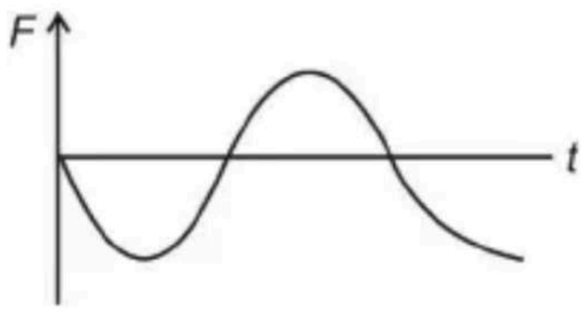
7. Displacement-time graph of a particle executing SHM is as shown in figure. The corresponding force-time graph of the particle can be



(1)


(2)


(3)


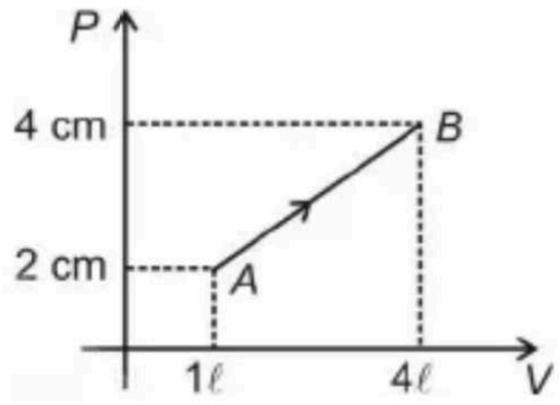
(4)


8. A U-tube containing a liquid moves with a horizontal acceleration a along the direction joining the two limbs. The separation between these two limbs is d. The difference in their liquid level is

(1) $\frac{ad}{g}$ (2) $\frac{2da}{g}$

(3) $\frac{da}{2g}$ (4) $d\tan\frac{a}{g}$

9. From the PV graph as shown in figure. What is the ratio of r.m.s. speed at B and A?



(1) 2 : 1 (2) 1 : 2

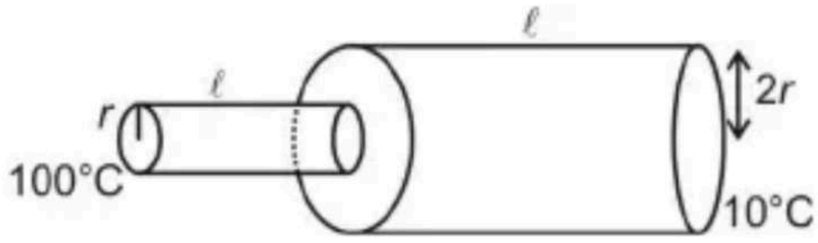
(3) $2\sqrt{2}:1$ (4) $1:\sqrt{2}$

10. 1 gm of ice at 0°C is mixed with 2 gm of water at 90°C. The final temperature of the mixture will be

(1) 20°C (2) $\frac{100}{3}$°C

(3) $\frac{40}{3}$°C (4) 30°C

11. Two wire A and B of same length different radius and different thermal conductivity are joint together as shown in figure. The temperature of the junction in thermal equilibrium will be



(1) 20°C (2) 25°C

(3) 35°C (4) 45°C

12. Specific heat of gas undergoing adiabatic change is

(1) Zero (2) Infinite

(3) Positive (4) Negative

13. A metal ball of surface area 200 cm^2, temperature 527°C is surrounded by a vessel at 27°C. If the emissivity of metal is 0.4, then net rate of loss of heat is nearly

($\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\ k^{-4}$)

(1) 108 W (2) 168 W

(3) 182 W (4) 192 W

14. A particle is projected from ground from origin. The position co-ordinates of particle varies with time as $x = 6t$ and $y = 8t - 5t^2$. The range of particles is

(1) 48 m (2) 4.8 m

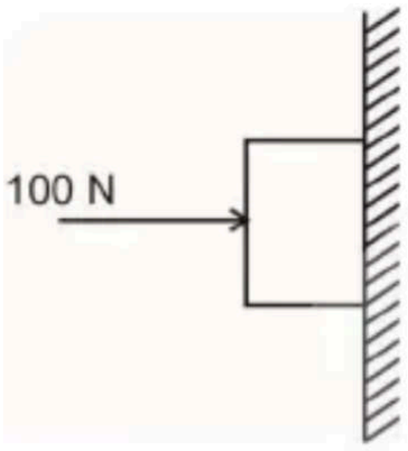
(3) 9.6 m (4) 24 m

15. If vector $\vec{A}$ and $\vec{B}$ have an angle θ between them, then value of $\left|\hat{A}-\hat{B}\right|$ will be

(1) $2\cos\frac{\theta}{2}$ (2) $2\tan\frac{\theta}{2}$

(3) $2\sin\frac{\theta}{2}$ (4) $2\cot\frac{\theta}{2}$

16. A horizontal force of 100 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2. The weight of the block is



(1) 20 N (2) 2 N

(3) 50 N (4) 5 N

17. Wind is blowing in north direction at the speed of 2 m/s which causes the rain to fall at some angle with the vertical. With what velocity should a cyclist drive so that the rain appears vertical to him

(1) 2 m/s South (2) 2 m/s North

(3) 4 m/s West (4) 4 m/s South

18. Kinetic energy of a particle moving in a straight line is proportional to the time t. The magnitude of force acting on the particle is

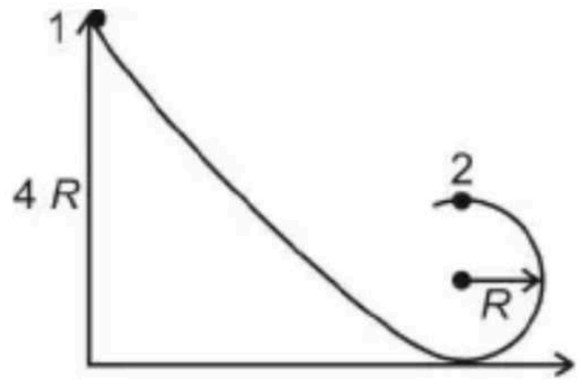
(1) Directly proportional to velocity of particle is

(2) Inversely proportional to $\sqrt{t}$

(3) Inversely proportional to speed of the particle

(4) Both (2) and (3)

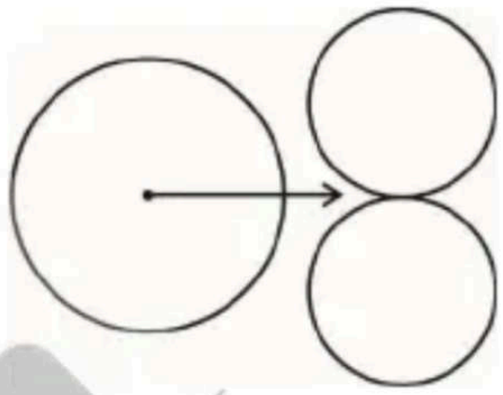
19. A point mass m starts from rest from point 1 at a height $4R$, where R is the radius of the circular track. The point mass slides down the frictionless track which ends in circular loop. The force which track exerts on the mass at point 2 is



(1) $3mg$ (2) mg

(3) $2mg$ (4) $4mg$

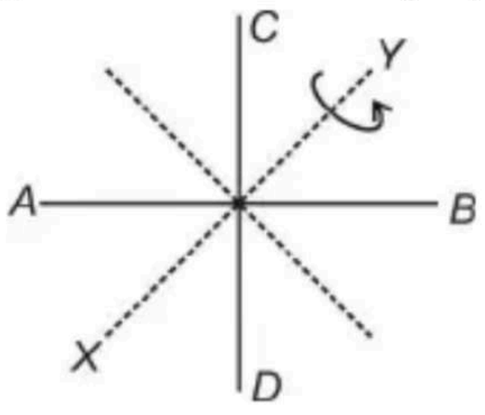
20. Two identical discs of equal radius and same mass are kept in contact on a frictionless table. A third disc of same mass but of double radius strikes them symmetrically and comes to rest after impact. The co-efficient of restitution is



(1) $\frac{2}{3}$ (2) $\frac{1}{3}$

(3) $\frac{9}{16}$ (4) $\frac{\sqrt{3}}{2}$

21. AB and CD are two identical rods each of length ℓ and mass m joined to form a cross. The moment of inertia of the system about a bisector of the angle between the rods (XY).



(1) $\frac{m\ell^2}{12}$ (2) $\frac{m\ell^2}{6}$

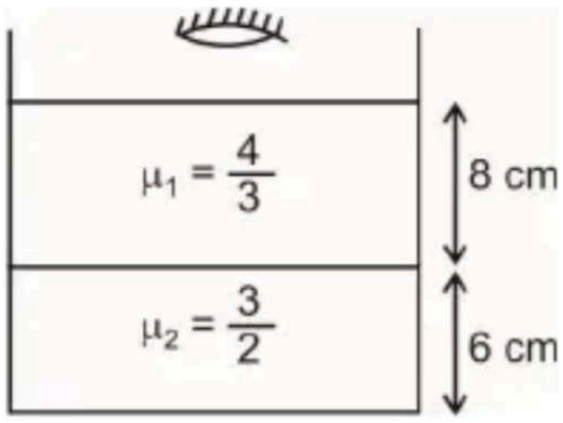
(3) $\frac{m\ell^2}{3}$ (4) $\frac{4m\ell^2}{3}$

22. A hollow sphere of mass m and radius r is rolling without slipping on a horizontal surface so that its centre of mass has a speed v. The work done required to stop the hollow sphere will be

(1) $\frac{7}{10}mv^2$ (2) $\frac{5}{6}mv^2$

(3) $\frac{3}{2}mv^2$ (4) $\frac{1}{2}mv^2$

23. In a vessel two insoluble liquids are filled as shown in figure. The effective refractive index of the combination for normal view is



(1) 1.33 (2) 1.5

(3) 1.67 (4) 1.4

24. A capacitor of capacitance C and resistance R are joined in series and connected to a source of voltage V and angular frequency ω. Power dissipated in the circuit will be

(1) $\left(R^2 + \frac{1}{\omega^2 C^2}\right)$ (2) $\frac{V^2 R}{\left(R^2 + \frac{1}{\omega^2 C^2}\right)}$

(3) $\frac{V}{R^2 + \frac{1}{\omega^2 C^2}}$ (4) $\frac{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}}{V^2}$

25. A diffraction pattern is obtained by using a beam of yellow light. What happens if the yellow light replaced by red light?

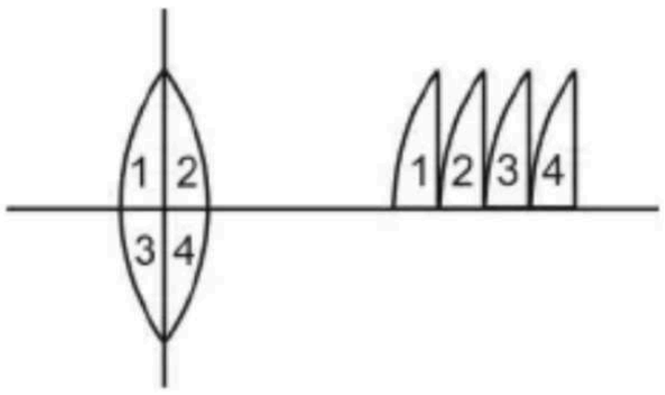
(1) No change

(2) Diffraction bands becomes narrower

(3) Diffraction band becomes broader

(4) Diffraction pattern disappears

26. A lens is cut from the optical axis into four equal parts. All parts of the lens are placed sideways as shown in the figure. Then new focal length of the combination



(1) Remains the same

(2) Becomes half

(3) Becomes $\frac{1}{4}$th times

(4) Becomes $\frac{1}{8}$th times

27. In Young's double slit experiment, fourth bright fringe is formed at a distance x from central fringe. The distance between it and second dark fringe on the same side of central maxima is

(1) $\frac{3x}{8}$ (2) $\frac{11x}{8}$

(3) $\frac{13x}{8}$ (4) $\frac{5x}{8}$

28. Unpolarized light of intensity I_0 passes through three polarizer such that transmission axes of the first and second polarizer makes angle 60°. With each other and the transmission axis of the last polarizer is crossed with that of the first. The intensity of the transmitted light will be

(1) $\frac{I_0}{16}$ (2) $\frac{I_0}{8}$

(3) $\frac{3I_0}{32}$ (4) $\frac{3I_0}{8}$

29. A proton and an electron are placed in a uniform electric field then

(1) Their acceleration will be equal

(2) The magnitude of their acceleration will be equal

(3) The electric forces acting on them will be equal

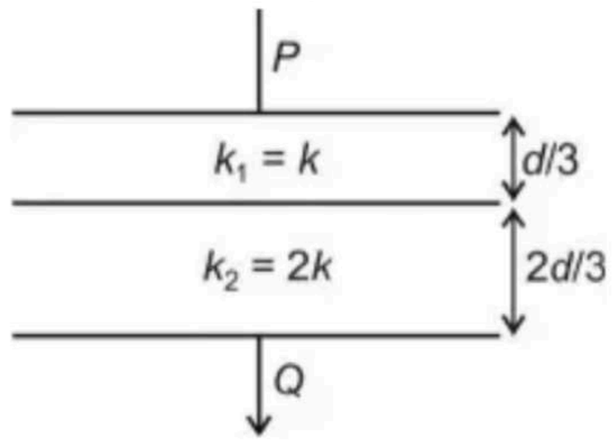
(4) The magnitude of the forces acting on them will be equal

30. An electric dipole of moment $\vec{P}$ placed in a uniform electric field $\vec{E}$ has maximum potential energy when the angle between $\vec{P}$ and $\vec{E}$ is

(1) Zero (2) $\frac{\pi}{2}$

(3) π (4) $\frac{3\pi}{2}$

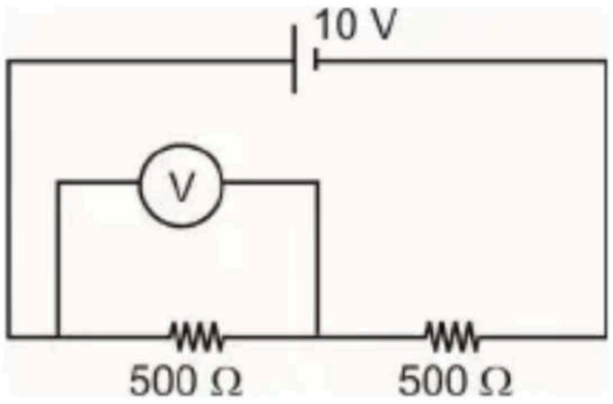
31. Two parallel plates of area A are separated by two dielectrics as shown in figure. The net capacitance between points P and Q is



(1) $\frac{3A\varepsilon_0 k}{2d}$ (2) $\frac{3A\varepsilon_0 k}{d}$

(3) $\frac{A\varepsilon_0 k}{2d}$ (4) $\frac{2A\varepsilon_0 k}{3d}$

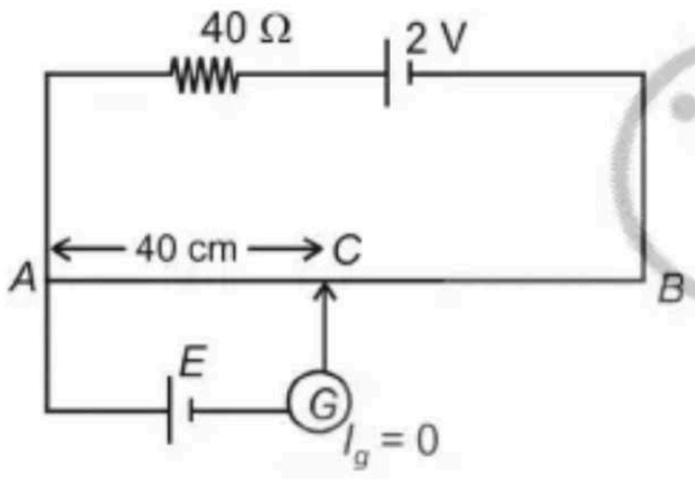
32. A voltmeter of resistance of 1000 Ω is connected across a resistance of 500 Ω in the given circuit. The reading of the voltmeter will be



(1) 1 V (2) 2 V

(3) 6 V (4) 4 V

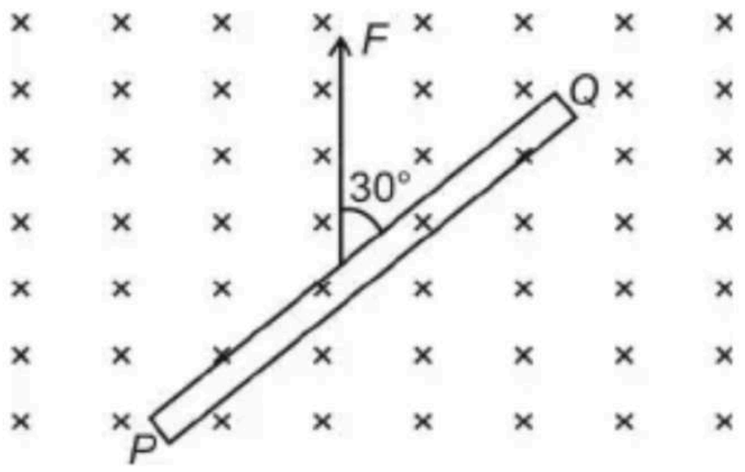
33. *AB* is potentiometer wire of length 100 cm and its resistance is 10 Ω, it is connected in series with a resistance 40 Ω and battery of emf 2 V and negligible internal resistance. If a battery of emf *E* is connected with potentiometer as shown in figure. The value of *E* will be



(1) 0.8 V (2) 1.6 V

(3) 0.08 V (4) 0.16 V

34. A conducting rod of mass *m* and length ℓ placed on smooth horizontal surface in a region where transverse uniform magnetic field *B* exist. At $t = 0$, a constant force *F* start acting on the rod at its mid point as shown. Potential difference between ends the rod, $V_P - V_Q$ at any time *t* is



(1) $\frac{BF\ell t}{2m}$ (2) $\frac{BF\ell t}{4m}$

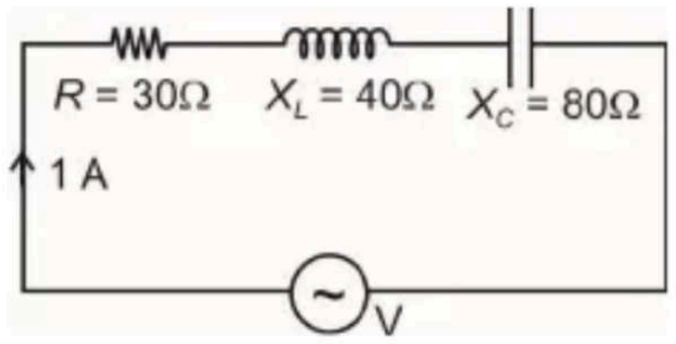
(3) $\frac{5BF\ell t}{8m}$ (4) $\frac{7BF\ell t}{8m}$

35. If a change in current of 0.01 A/s in one coil produces a change in magnetic flux of 2×10^{-2} W b/s in other coil, then the mutual inductance of two coils in henry will be

(1) Zero (2) 0.5

(3) 2 (4) 3

36. In series LCR circuit as shown in figure. Value of *V* will be



(1) 50 V (2) 100 V

(3) 30 V (4) 40 V

37. For photoelectric emission from a certain metal, cut off frequency is ν. If radiation of frequency 2ν impinges on the metal plate, the maximum possible velocity of the emitted electron will be (*m* is mass of the electron)

(1) $2\sqrt{\frac{2h\nu}{m}}$ (2) $\sqrt{\frac{2h\nu}{m}}$

(3) $\sqrt{\frac{h\nu}{2m}}$ (4) $\sqrt{\frac{h\nu}{m}}$

38. The nuclei of which of the following pairs of nuclei are isotone?

(1) $_{34}Sc^{74}$, $_{31}Ca^{71}$ (2) $_{32}Mo^{92}$, $_{40}Zr^{92}$

(3) $_{38}Sr^{81}$, $_{38}Sr^{86}$ (4) $_{20}Cs^{40}$, $_{16}S^{32}$

39. A radioactive substance has a half-life 1 year. The fraction of this material, that would remain after 4 years will be

(1) $\frac{1}{32}$ (2) $\frac{1}{4}$

(3) $\frac{1}{16}$ (4) $\frac{1}{8}$

40. The radius of electrons second stationary orbit in Bohr's atom is *R*. The radius of fourth orbit will be

(1) 2*R* (2) *R*

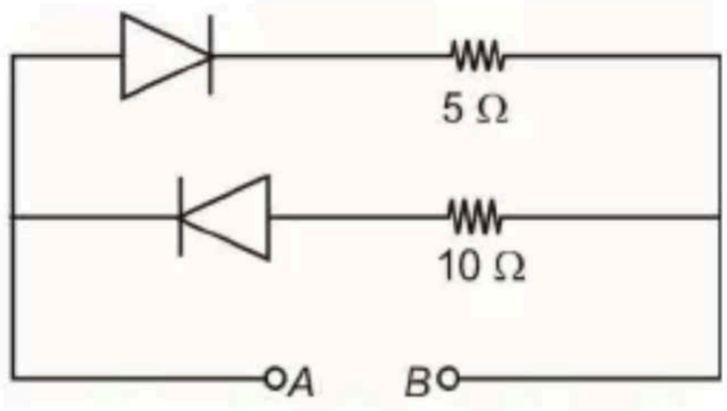
(3) 8*R* (4) 4*R*

41. The ripple factor of a half wave rectifier is

(1) 0.48 (2) 0.6

(3) 1.51 (4) 1.21

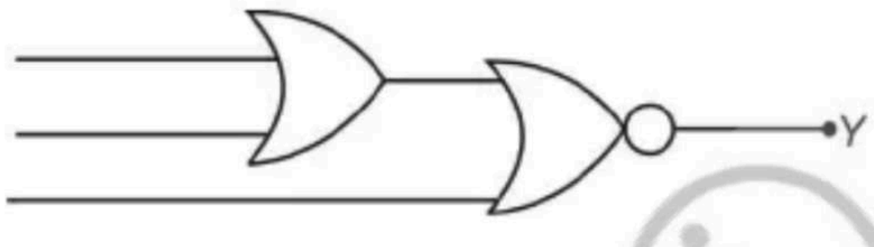
42. A 4 V battery is connected across the point *A* and *B* as shown in the figure given below. Assuming that the resistance of each diode is zero in forward bias and infinity in reverse bias. The current supplied by the battery when the positive terminal is connected to *A* will be



(1) 0.4 A (2) 0.6 A

(3) 0.8 A (4) 0.2 A

43. The output *y*, when all three inputs are first high then low will be



(1) 1, 1

(2) 0, 0

(3) 0, 1

(4) 1, 0

44. A diverging lens is dipped in a liquid whose refractive index is more than the refractive index of material of lens then

(1) It behave like a glass slab

(2) It still behave like a diverging lens

(3) It behave like a converging lens

(4) None of these

45. A straight wire carrying current *I* is made into a square loop. If *M* is the magnetic moment associated with the loop, then length of the wire will be

(1) $2\sqrt{\frac{M}{I}}$ (2) $4\sqrt{\frac{M}{I}}$

(3) $\sqrt{\frac{M}{I}}$ (4) $\sqrt{\frac{2M}{I}}$

CHEMISTRY

46. If the heat released during the neutralization reaction $NaOH + HNO_3 \longrightarrow NaNO_3 + H_2O$ is 57.1 kJ/mol, then the heat released when 0.1 mole of NaOH neutralizes 0.1 mole of HNO_3 is

(1) 5.71 kJ (2) 57.1 kJ

(3) 0.57 kJ (4) 28.5 kJ

47. Enthalpy of reaction for the following reaction

$CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$ is

(Bond Energy of C – H = 414 kJ/mol

Cl – Cl = 243 kJ/mol

C – Cl = 331 kJ/mol

H – Cl = 431 kJ/mol)

(1) 105 kJ (2) – 105 kJ

(3) 209 kJ (4) – 209 kJ

48. pH of 0.2 M sodium acetate solution is (pK_a of CH_3COOH is 4.74)

(1) 8.5 (2) 11

(3) 9.7 (4) 9

49. Which among the following is/are negatively charged sol(s)?

(1) TiO_2 sol (2) Copper sol

(3) Gelatin sol (4) Both (2) and (3)

50. A gas has a vapour density of 5.6. The volume occupied by 10 g of the gas at STP will be

(1) 5 L (2) 20 L

(3) 10 L (4) 30 L

51. The temperature at which the average speed of N_2 molecules is same as that of O_2 molecules at 1327°C is

(1) 1200 K (2) 2800 K

(3) 1400 K (4) 1800 K

52. Which of the following solutions will exhibit the highest freezing point?

(1) 0.1 M NaCl (2) 0.1 M glucose

(3) 0.1 M $MgSO_4$ (4) 0.1 M $AlCl_3$

53. The wavelength of visible light varies from

(1) 300 nm to 650 nm

(2) 400 nm to 750 nm

(3) 1050 nm to 1500 nm

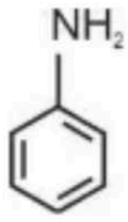
(4) 750 nm to 1050 nm

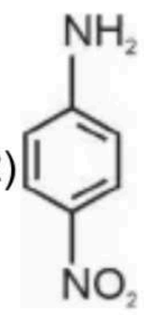
54. Maximum number of electrons that can be associated with the set of quantum numbers $n = 4, l = 2, m = -1, s = +\frac{1}{2}$, is

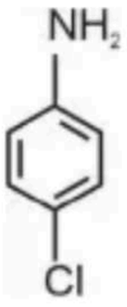
(1) 1 (2) 2

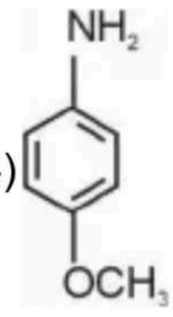
(3) 4 (4) 6

55. Which of the following undergoes diazotization most readily?

(1) 

(2) 

(3) 

(4) 

56. 68.4 g cane sugar and 36 g glucose are dissolved in 2 L water at 300 K. The osmotic pressure of the solution will be

(1) 8.7 atm (2) 12.1 atm

(3) 4.9 atm (4) 3.2 atm

57. In CaF_2, fluorite type structure, F^- ions occupy

(1) All the tetrahedral voids

(2) Alternate tetrahedral voids

(3) All the octahedral voids

(4) All the octahedral voids + half of tetrahedral voids

58. A metal crystallises in bcc unit cell. If the distance between its nearest atoms is 1.8 Å, then the edge length of the unit cell is

(1) 1.2 Å (2) 2.4 Å

(3) 2.07 Å (4) 1.8 Å

59. In which of the following options, order of arrangement does not agree with the variation of property indicated against it?

(1) I < Br < F < Cl (negative electron gain enthalpy)

(2) Be < C < B < Li (Second ionisation enthalpy)

(3) $Ba(OH)_2 < Sr(OH)_2 < Ca(OH)_2 < Mg(OH)_2$ (Basic nature of hydroxides)

(4) C < N < O < F (electronegativity)

60. Assuming *s-p* mixing is not operative, the paramagnetic species among the following is

(1) N_2

(2) O_2

(3) B_2

(4) Both (2) and (3)

61. Consider the following reaction

$A + 4B + 3C + 2D \longrightarrow 5E + 3F$

If 4 mole of A, 14 mole of B, 6 mole of C reacts with 6 mole of D, then the limiting reagent is

(1) A (2) B

(3) C (4) D

62. The maximum number of electrons present in Cr^{3+} having value of (n + l) = 3 is

(1) 4 (2) 6

(3) 8 (4) 10

63. Which of the following is a mineral of copper?

(1) Siderite (2) Azurite

(3) Cryolite (4) Calamine

64. In which of the following reactions, PH_3 is not formed in the product?

(1) $Ca_3P_2 + H_2O \longrightarrow$

(2) $PCl_3 + H_2O \longrightarrow$

(3) $P_4 + NaOH + H_2O \longrightarrow$

(4) $PH_4I + KOH \longrightarrow$

65. Which of the following is an incorrect statement?

(1) α-sulphur is yellow in colour while β-sulphur is colourless

(2) Specific gravity of α-sulphur is more than β-sulphur

(3) S_2 is diamagnetic

(4) Cyclo-S_6 has the chair form structure

66. The electronic configuration of La(57), Lu(71) and Cm(96) respectively are

(1) $[Rn]5f^1\ 7s^2$, $[Xe]\ 4f^{14}\ 5d^1\ 6s^2$, $[Rn]\ 5f^9\ 7s^1$

(2) $[Rn]5f^1 7s^2$, $[Xe]\ 4f^{14} 6s^2$, $[Rn]\ 5f^7\ 6d^1\ 7s^2$

(3) $[Xe]5d^1\ 6s^2$, $[Xe]\ 4f^{14}5d^1\ 6s^2$, $[Rn]\ 5f^7\ 6d^1\ 7s^2$

(4) $[Xe]4f^1\ 6s^2$, $[Xe]\ 4f^{10}\ 6d^5\ 6s^2$, $[Rn]\ 5f^6\ 7s^2$

67. The correct order of third ionisation enthalpy is

(1) Zn > Sc > Mn > Fe (2) Zn > Fe > Mn > Sc

(3) Zn > Mn > Fe > Sc (4) Mn > Zn > Fe > Sc

68. The energy of an octahedral high spin complex with d^7 configuration of cation will be (CFSE in an octahedral complex = Δ_0, Electron pairing energy = P)

(1) $\frac{-3}{4}\Delta_0 + 2P$ (2) $\frac{-4}{5}\Delta_0 + 2P$

(3) $\frac{-9}{5}\Delta_0 + 3P$ (4) $-\frac{1}{5}\Delta_0 + P$

69. Which one of the following is not a common component of classical smog?

(1) Smoke (2) Fog

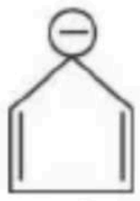
(3) Sulphur dioxide (4) O_3

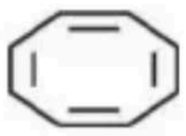
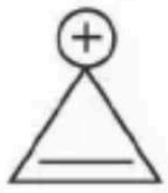
70. Which of the following groups deactivates the benzene ring towards electrophilic substitution reaction?

(1) –OH (2) $-NO_2$

(3) $-NH_2$ (4) –OR

71. Non-aromatic species among the following is

(1)  (2) 

(3)  (4) 

72. During estimation of nitrogen present in an organic compound by Kjeldahl's method, the ammonia evolved from 0.1 g of the compound is neutralized by 5 mL of 0.05 M H_2SO_4. The percentage of nitrogen in the compound is

(1) 7% (2) 14%

(3) 35% (4) 60%

73. The formula of calgon is

(1) $Na_2[Na_4(PO_3)_6]$ (2) Na_2CO_3

(3) $Na_2P_4O_{18}$ (4) $Na_2Al_2Si_2O_8$

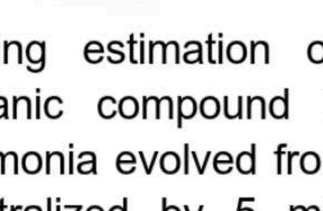
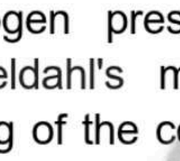
74. Which among the following is a purine base?

(1) Thymine (2) Uracil

(3) Adenine (4) Cytosine

75. The compound that will give Tollen's test but will not give iodoform test is

(1) CH_3CHO (2)

(3)  (4) 

76. Correct order of reactivity of carboxylic acids towards esterification reaction is

(1) $HCOOH < CH_3COOH < CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COOH$

(2) $CH_3COOH < H-COOH < CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COOH$

(3) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COOH < CH_3COOH < HCOOH$

(4) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COOH < HCOOH < CH_3COOH$

77. Iproniazid is an example of

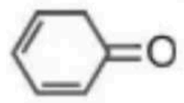
(1) Tranquilizer (2) Antimicrobial

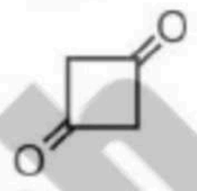
(3) Analgesic (4) Antiseptic

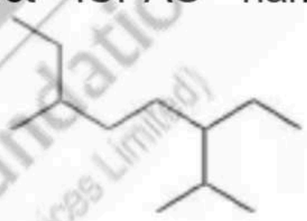
78. Among the following, maximum enol content is present in

(1) $C_6H_5-\overset{\displaystyle O}{\overset{\|}{C}}-CH_2-\overset{\displaystyle O}{\overset{\|}{C}}-CH_3$

(2) $CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-CH_2-\overset{\displaystyle O}{\overset{\|}{C}}-O-C_2H_5$

(3) 

(4) 

79. The correct IUPAC name of the following compound  is

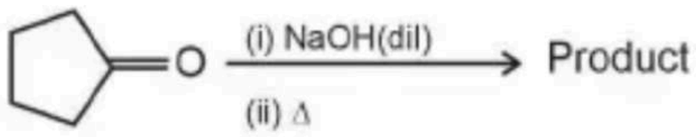
(1) 6-methyl-3-(methyl ethyl) octane

(2) 3,6-diethyl-2-methylheptane

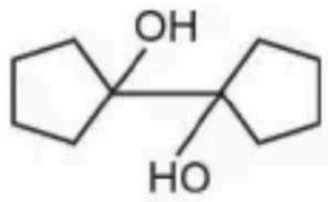
(3) 2,6-dimethyl-3-ethyloctane

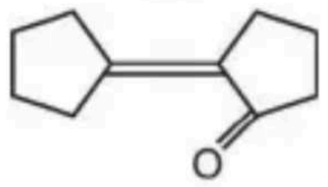
(4) 3-ethyl-2,6-dimethyloctane

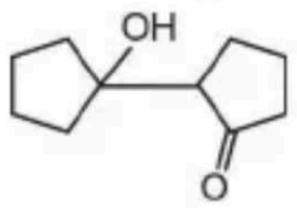
80. In the reaction,

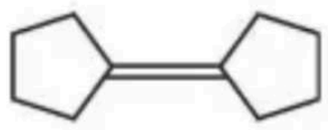


Major product is

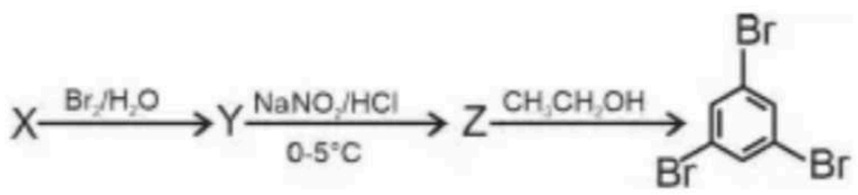
(1) 

(2) 

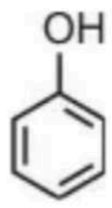
(3) 

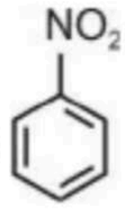
(4) 

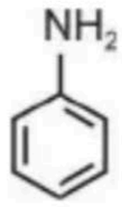
81. In the reaction sequence,



Identify X

(1) 

(2) 

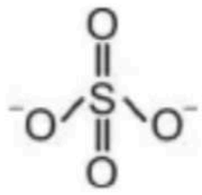
(3) 

(4) 

82. Expanded octet is present in

(1) CH_4 (2) $AlCl_3$

(3) SF_6 (4) NH_3

83. Formal charge on S in  is

(1) Zero (2) +1

(3) +2 (4) –2

84. In which of the following molecules both polar and non-polar bonds are present?

(1) H_2O_2 (2) PCl_5

(3) ClF_5 (4) XeO_3

85. For which of the following, standard enthalpy of formation is zero?

(1) C(g) (2) $Br_2(g)$

(3) C(graphite) (4) C(Diamond)

86. If the enthalpy and entropy change for the reaction $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$ are 75 kJ mol^{-1} and 250 JK^{-1} mol^{-1} respectively. The temperature at which the reaction will be at equilibrium is

(1) 250 K (2) 300 K

(3) 400 K (4) 450 K

87. Half life of a first order reaction is 6.93 min. Percentage of reaction that will be completed in 46.06 min. will be

(1) 99% (2) 90%

(3) 85% (4) 95%

88. For one mole of a monoatomic gas γ is equal to

(1) 1.66 (2) 1.40

(3) 1.33 (4) 1.27

89. Which of the following is incorrect for a cyclic process?

(1) $\Delta H = 0$ (2) $\Delta S = 0$

(3) $\Delta U = 0$ (4) $W = 0$

90. Monomer of orlon is

(1) Vinyl chloride (2) Ethene

(3) Acrylonitrile (4) Tetrafluoroethene

BOTANY

91. Which of the following is **correct** option for sequence of taxonomic categories?

(1) Species → Family → Genus

(2) Species → Order → Genus

(3) Kingdom → Order → Phylum

(4) Species → Genus → Family

92. Select the **correct** statement(s) about metabolism.

(1) Substances are broken down to release energy

(2) Substances are synthesised by gain of energy

(3) Occurs only in multicellular organisms

(4) Both (1) and (2)

93. Bacteria involved in the conversion of N_2 into NH_3 are

(1) *Nitrococcus* (2) *Nitosomonas*

(3) *Azotobacter* (4) *Nitrobacter*

94. Smallest organisms which are self-reproducing independently are

(1) Viruses (2) Prion

(3) Bacteria (4) PPLO

95. Kingdom Monera includes all, **except**

(1) Methenogens

(2) Thermophillic bacteria

(3) Halophiles

(4) Protozoans

96. Fungal cell wall is composed of

(1) Peptidoglycan (2) Pectin

(3) Chitin (4) Both (2) and (3)

97. Which of the following is true for fungi imperfecti?
(1) They do not have sexual phase
(2) They include toad stools
(3) They include *Aspergillus*
(4) They are known as sac fungi

98. Red tides are caused by
(1) Diatoms (2) Cyanobacteria
(3) Brown algae (4) Dinoflagellates

99. Which of the following is common bread mould?
(1) *Aspergillus* (2) *Penicillium*
(3) Yeast (4) *Rhizopus*

100. Zygotic meiosis occurs in
(1) Angiosperms
(2) Vascular cryptogams
(3) Gymnosperms
(4) Algae

101. Vascular tissues are **not** present in
(1) Gymnosperms (2) Angiosperms
(3) Bryophytes (4) Pteridophytes

102. Alga which produces agar agar is
(1) *Nostoc* (2) *Volvox*
(3) *Gracilaria* (4) *Anabaena*

103. Male sex organs of bryophytes are
(1) Oogonia (2) Ascospores
(3) Archegonia (4) Antheridia

104. Juvenile stage of mosses is called
(1) Capsule (2) Archegonia
(3) Prothallus (4) Protonema

105. Dead fibre associated with the food translocating vascular tissues in plants are
(1) Parenchyma (2) Wood fibre
(3) Bast fibre (4) Collenchyma

106. In angiosperms, phloem is composed of
(1) Tracheids and vessels
(2) Vessels and fibre
(3) Sieve elements and companion cells
(4) Sieve cells and fibre

107. Vascular bundles are scattered in stem of all the following **except**
(1) Wheat (2) Maize
(3) Rice (4) Gram

108. Cork cambium in dicot root is derived from
(1) Hypodermis (2) Pericycle
(3) Pith (4) Epidermis

109. Exchange of genetic material between homologous chromosomes occurs in
(1) Telophase-I (2) Prophase-I
(3) Anaphase-II (4) Prophase-II

110. Segregation of Mendelian factors occurs during
(1) Anaphase-I (2) Telophase-II
(3) Zygotene (4) Prophase-II

111. Detoxification of toxic substances are done by
(1) RER (2) SER
(3) Lysosomes (4) Golgi apparatus

112. All of the following are bounded by membrane **except**
(1) Chloroplast (2) Mitochondria
(3) Lysosome (4) Centrioles

113. Difference between OP and TP is known as
(1) Suction pressure (2) Transipration pull
(3) Osmotic potential (4) Solute potential

114. Stomata opening occurs mainly due to
(1) Increased turgor pressure of the guard cells.
(2) Increased concentration of CO_2 in the sub stomatal cavity.
(3) Increased ABA concentration.
(4) High potassium concentration in subsidiary cell

115. During photorespiration, oxygen is consumed in
(1) Chroloplast
(2) Mitochondria
(3) Peroxisomes
(4) Both chloroplast and mitochondria

116. CO_2 fixation occurs in C_4 plants even at low CO_2 concentration because
(1) RuBisCO in C_4 plants is very sensitive for CO_2 in mesophyll cells
(2) Primary fixation of CO_2 occurs in bundle sheath cells
(3) PEPcase has higher affinity for CO_2
(4) There is CO_2 pump found in cells

117. First enzyme of EMP pathway is
(1) Pyruvate kinase
(2) Hexokinase
(3) Citrate synthase
(4) Phosphofructokinase

118. Which of the following is final electron acceptor in ETS during aerobic respiration?
(1) CO_2 (2) H_2O
(3) O_2 (4) NADH

119. Which of the following is used as weedicide as well as shows apical dominance?
(1) Ethylene (2) ABA
(3) 2, 4-D (4) GAs

120. **A.** Auxins supress growth of lateral buds.
B. GAs are responsible for bolting in plants.
Select the **correct** option.
(1) Both A and B are incorrect
(2) Both A and B are correct
(3) Only A is correct
(4) Only B is correct

121. An egg apparatus in angiosperms is composed of
(1) An egg cell and two antipodal cells
(2) An egg cells and secondary nucleus
(3) Two egg cells and PEN
(4) An egg cell and two syngergids

122. Remnant of nucellus present in the seeds of
(1) Beet (2) Wheat
(3) Rice (4) Maize

123. In a DNA molecule, adenine is 35%. What would be the percentage of cytosine in DNA?
(1) 10% (2) 20%
(3) 15% (4) 30%

124. Given mRNA codes for how many amino acids?
5′ AUGUGAAUUUAA 3′ ?
(1) One (2) Two
(3) Three (4) Four

125. RNA formed from the transcription of DNA has base sequence 5′ AUUAUGAAA 3′ . What would be the sequence of coding strand of DNA?
(1) 5′ AUUAUGAAA 3′ (2) 3′ UAAUACUUU 5′
(3) 5′ ATTATGAAA 3′ (4) 3′ AUUAUGAAA 5′

126. Phenylketonuria can be characterised by the
(1) Defected haemoglobin
(2) Sickle shaped RBCs
(3) Mental retardation
(4) High concentration of urea in urine

127. Sex linked trait in human is
(1) Sickle cell anemia (2) Phenylketonuria
(3) Down's syndrome (4) Color blindness

128. In which of the following CO_2 is not the last product?
(1) Alcoholic fermentation
(2) Lactic acid fermentation
(3) Aerobic respiration in cyanobacteria
(4) Anaerobic respiration in yeast

129. Which of the following organisms produce the immunosuppressive agent cyclosporin A?
(1) *Pseudomonas* (2) *Aspergillus*
(3) *Clostridium* (4) *Trichoderma*

130. Zooplanktons found in oceans are
(1) Primary producers
(2) Primary consumers
(3) Secondary consumer
(4) Saprophytes

131. In nitrification, ammonia is converted into NO_2^- by
(1) *Nitrobacter* (2) *Pseudomonas*
(3) *Nitrosomonas* (4) *Azotobacter*

132. Which of the following is **not** the control measure of global warming?
(1) Planting trees
(2) Slowing down population growth
(3) Reducing deforestation
(4) More use of chloroflurocarbons

133. An association between two organisms in which one is benefitted while the other is not affected at all is known as
(1) Commensalism (2) Ammensalism
(3) Symbiosis (4) Cooperation

134. Water blooms caused by water hyacinth is mainly due to
(1) Biomagnification
(2) High salts in water body
(3) Eutrophication
(4) Less organic material in water body

135. BOD is an indication of
(1) Pure water inflow in water body
(2) High CO_2 in water body
(3) High organic material in water body
(4) High O_2 present in a lake

ZOOLOGY

136. Which junction helps to stop substances from leaking across a tissue?

(1) Tight junction (2) Gap junction

(3) Adhering junction (4) Desmosomes

137. Read following statements w.r.t connective tissue and choose the option representing number of **correct** statements.

a. Connective tissue is the most abundant and widely distributed tissue in the body of complex animals.

b. Tendons and ligaments are specialized connective tissues in our body

c. Blood is a fluid connective tissue containing plasma and fibres

d. Adipose tissue is located beneath the skin

(1) 3 (2) 4

(3) 2 (4) 1

138. During oogenesis in humans, second meiotic division is completed

a. After puberty

b. After ovulation

c. Before fertilization

d. During maturation phase

e. Before karyogamy during fertilization

Choose the option with all **correct** statements.

(1) a, b and c

(2) a, b, d and e

(3) b, c, d and e

(4) a, b, c, d and e

139. Select the **correct** statement w.r.t female reproductive system.

(1) Ovary is also known as womb

(2) Myometrium of uterus is responsible for mechanical function

(3) Clitoris lies at lower junction of two labia minora

(4) Opening of cervix is covered by hymen

140. The most vital event of sexual reproduction is

(1) Gametogenesis

(2) Gamete transfer

(3) Syngamy

(4) Cleavage of zygote

141. Select **incorrect** statement w.r.t sexual reproduction

(1) Sexual reproduction involves formation of male and female gametes always in different individuals of same species.

(2) It is an elaborate and complex process in comparison to asexual reproduction.

(3) Offsprings are not completely identical to parents

(4) Due to variations, it is significant for evolution.

142. A healthy human ejaculates sperms in semen with sperm count 80 million sperms per ml for normal fertility. How many sperms must have normal shape, size and vigorous motility per ejaculate of 3 ml semen?

(1) 144 million (2) 72 million

(3) 57.6 million (4) 120 million

143. Read the following statements A and B and choose the **correct** option.

Statement-A : Lactational amenorrhoea has been reported to be effective upto a period of six months following parturition.

Statement-B : Side effects are almost nil and failure rate is high in case of natural methods of contraception.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

144. Which of the following is a hormone releasing IUD?

(1) Lippes loop (2) CuT

(3) Multiload - 375 (4) LNG - 20

145. The natural selection in which more individuals acquire a value other than mean character value is called

(1) Disruptive selection (2) Directional selection

(3) Stabilizing selection (4) Sexual selection

146. All of the following are evolved from progymnosperms **except**

(1) Seed ferns (2) Conifers

(3) Cycads (4) Dicotyledons

147. Which of the following is **incorrect** statement w.r.t evolutionary history of vertebrates?

(1) Mammals are evolved directly from therapsids

(2) Birds are phylogenetically closer to dinosaurs than crocodiles

(3) Snakes are evolved with turtles in Jurassic period

(4) Pelycosaurs are evolved from synapsids

148. Select **odd** one w.r.t substances and their categories/functions.

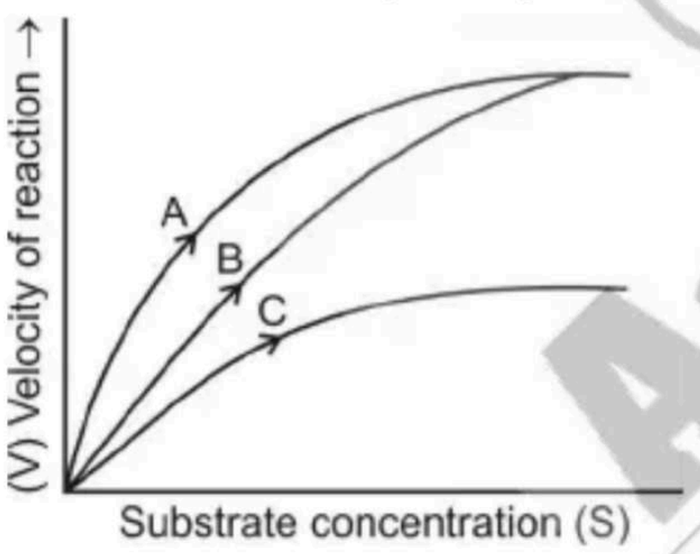
(1) Trypsin – Enzyme

(2) GLUT-4 – Enables glucose transport into cells

(3) Alkaloid – Codeine

(4) Pigment – Cellulose

149. The graphical representation of enzyme reaction is shown in three different conditions. The graph is plotted between velocity of reaction at y-axis and substrate concentration at x-axis. What do the curve A, B and C depict respectively?



(1) A-normal enzyme reaction, B-competitive inhibition C-non-competitive inhibition.

(2) A- enzyme with allosteric modulation, B-normal enzyme activity, C-competitive inhibition.

(3) A-enzyme with optimum temperature, B-non-competitive inhibition, C-competitive inhibition.

(4) A-normal enzymatic reaction, B-non-competitive inhibition, C-allosteric inhibition.

150. Widal test is used for diagnosis of

(1) Malaria

(2) Typhoid

(3) Ascariasis

(4) AIDS

151. 'Smack' is a drug obtained from

(1) Latex of *Papaver somniferum*

(2) Leaves of *Cannabis sativa*

(3) Flowers of *Dhatura pinata*

(4) Fruits of *Erythroxylum*

152. Which of the following statements is **correct** w.r.t AIDS?

(1) The HIV enters helper T-lymphocytes thus reducing their numbers.

(2) The HIV is always transmitted through eating food together with an infected person.

(3) Drug addicts are least susceptible to HIV infection.

(4) AIDS patients are being fully cured with proper care and nutrition.

153. Read following statements A to D and choose the option which **correctly** represents true and false statements.

A. Saliva in buccal cavity and HCl in stomach are physiological barriers of the body included in innate immunity.

B. Immunoglobulins are glycoproteins in which both light and heavy chains are linked together by intrachain disulphide bridges.

C. IgM are largest antibodies which participate in primary immune response.

D Malignant tumors are invasive, metastatic and show property of contact inhibition.

	A	B	C	D
(1)	True	True	False	False
(2)	True	False	False	True
(3)	True	False	True	False
(4)	False	False	True	True

154. Select the **incorrect** statement w.r.t MOET

(1) MOET has been demonstrated for cows, sheep, mare etc.

(2) MOET is used to increase herd size in short time.

(3) The fertilized eggs at 8-32 celled stages are recovered non surgically and transferred to genetic mother.

(4) In this method, a cow is administered hormones, with FSH-like activity.

155. Read the following statements carefully and select the option with all **correct** statements.

a. The source of restriction enzyme *Bam* HI is *E. coli*

b. In biolistic method of gene transfer, microparticles made up of gold or tungsten are coated with foreign DNA.

c. Microinjection method for injecting recombinant DNA is used for animal cell.

d. Primers are chemically synthesized oligonucleotides that are complementary to the region of DNA in PCR.

(1) a, b, c and d (2) a, b and c

(3) a, c and d (4) b, c and d

156. GEAC stands for

(1) Genome engineering action committee

(2) General engineering action committee

(3) Genetic engineering approval committee

(4) Genetic and environmental assisted committee

157. Bt. toxin causes death of insect by causing

(1) Damage in brain

(2) Lysis of midgut epithelial cells

(3) Change in gut pH from alkaline to acidic

(4) Mouth parts to become non-functional

158. An improved variety of transgenic basmati golden rice

(1) Gives high yield and is rich in β-carotene which is transformed to vit-A in our body

(2) Is completely resistant to all insect pests and diseases of paddy

(3) Gives high yield but has no characteristic aroma

(4) Does not require chemical fertilizers.

159. Which of the following cells is **not** correctly matched with its secretion?

(1) Oxyntic cells – Castle's intrinsic factor

(2) Peptic cells – Secrete HCl

(3) Goblet cells – Secrete mucus

(4) Chief cells – Secrete pepsinogen

160. Dental formula of a normal adult human is

(1) $\frac{2123}{2123}$ (2) $\frac{2033}{1023}$

(3) $\frac{2102}{2102}$ (4) $\frac{2121}{2123}$

161. Which of the following situations would result in the greatest degree of O_2 saturation for haemoglobin, assuming pO_2 remains constant?

(1) Increased H^+, decreased temperature

(2) Increased CO_2 levels, decreased temperature

(3) Decreased CO_2 levels, decreased acidity

(4) Increased CO_2 levels, increased temperature

162. Read following statements carefully and select the option containing all **correct** statements.

a. pO_2 of atmospheric air is more than pO_2 of alveolar air.

b. Diffusion of gases through respiration membrane depends upon their solubility and partial pressure.

c. Intrapleural pressure during normal inspiration is less than intra pulmonary pressure.

d. Decrease in activity / frequency of pneumotaxic centre increases breathing rate.

(1) a, b and c (2) a, b and d

(3) b, c and d (4) a, c and d

163. Maximum forceful expiration after normal inspiration is called

(1) Vital capacity (2) Inspiratory capacity

(3) Expiratory capacity (4) Residual volume

164. Which of the following statements is **correct**?

(1) Cardiac output of an athlete is much lower than that of a normal man.

(2) A single cardiac cycle is performed per minute.

(3) Auriculoventricular valves and semilunar valves remains open during isovolumetric diastole.

(4) Cardiac cycle includes auricular systole, ventricular systole and joint diastole/ diastasis.

165. Which of the following WBCs participate directly in phagocytosis of foreign substances in blood?

(1) Neutrophils (2) Eosinophils

(3) Macrophages (4) Lymphocytes

166. Which of the following part of nephron has maximum obligatory water reabsorption?

(1) DCT (2) Loop of Henle

(3) PCT (4) Collecting duct

167. The amount of urea excreted out per day in a normal individual is about

(1) 25–30 mg

(2) 25–30 gm

(3) 20 mg

(4) 10–15 mg

168. All of the following are high in red muscle fibres **except**

(1) Myoglobin

(2) Mitochondria

(3) Large amount of O_2

(4) Sarcoplasmic reticulum

169. Each ____x____ has alternate dark and light bands. The light bands contain ____y____ and is called ____z____.

Fill in the blanks (x, y and z) with suitable options and choose the **correct** option.

	x	y	z
(1)	Muscle fibre	Myosin	A-band
(2)	Myofibril	Actin	I-band
(3)	Muscle fibre	Actin	A-band
(4)	Myofibril	Myosin	I-band

170. Match the column-I with column-II and choose the **correct** option.

Column-I	Column-II
a. Axoplasm	(i) Impermeable to -ve charge proteins
b. Axolemma	(ii) Low K^+ and high Na^+
c. Fluid outside axon	(iii) Active transport
d. $Na^+ – K^+$ pump	(iv) High K^+ and low Na^+

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(ii), c(iii), d(i)

(3) a(iii), b(ii), c(iv), d(i)

(4) a(iv), b(i), c(ii), d(iii)

171. Which of the following is **not** a part of reflex arc?

(1) Sensory neuron

(2) CNS

(3) Motor neuron

(4) Tissue fluid

172. Membranous labyrinth is filled with a fluid called ____A____ and surrounded by a fluid called ____B____ present inside ____C____.

Choose the option which represents A, B and C **correctly**.

	A	B	C
(1)	Perilymph	Endolymph	Cochlea
(2)	Perilymph	Endolymph	Bony labyrinth
(3)	Endolymph	Perilymph	Bony labyrinth
(4)	Perilymph	Cortilymph	Membranous labyrinth

173. The arrangement of retinal layer from outside to inside is

(1) Ganglionic layer, choroid and retina

(2) Photoreceptor cells, bipolar cells and ganglionic cells

(3) Bipolar cells, photoreceptor cells, ganglionic cells

(4) Layer of rods and cones, ganglionic cells, bipolar cells

174. Which of the following hormones are related with osteoporosis?

(1) Progesterone and aldosterone

(2) Estrogen and aldosterone

(3) Estrogen and parathormone

(4) Parathormone and prolactin

175. Steroid hormones

(1) Have only cell surface receptors

(2) Are produced only within adrenal cortex

(3) Act through receptors present within the cell

(4) Are lipophobic in physical property

176. Metameric segmentation and open type of circulatory system with jointed appendages is present in animals of phylum

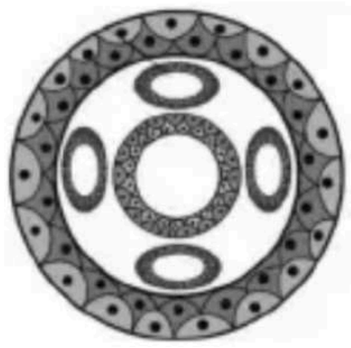
(1) Annelida

(2) Arthropoda

(3) Chordata

(4) Mollusca

177. There is a diagrammatic sectional view of an organism in following given figure.



The animals may be

(1) *Schistosoma* and *Ascaris*

(2) *Nereis* and *Trichinella*

(3) *Ascaris* and *Trichinella*

(4) *Fasciola* and *Taenia*

178. Choose the **incorrect** match with respect to functions of structures found in cockroach.

	Structure	**Function**
(1)	Phallic gland	Forms covering of spermatophore
(2)	Seminal vesicle	Storage of sperms
(3)	Forewings	Help in flight
(4)	Collaterial gland	Secrete hard egg-case oothecal

179. Which of the following statements is **incorrect**?

(1) Cyclostomes are fishes without paired fins and scales placed in class pisces.

(2) Birds have two additional chambers crop and gizzard in their alimentary canal.

(3) Crocodiles are cold blooded animals with diaphragm and four chambered heart.

(4) Presence of mammary glands is a feature of mammals without any exception.

180. Diploblastic animals without tentacles, having blind sac body plan showing bioluminescence are placed in phylum

(1) Porifera

(2) Coelenterate

(3) Ctenophora

(4) Mollusca

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Edition: 2020-21

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